

MAUVE-MUSE: A Resolved View of Star Formation, Gas Cycling, and Chemical Enrichment in Virgo Cluster Galaxies

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Event: CGER Galaxy Forum, Yonsei University

Schedule: 22 September 2026, 16:00, Room 638, Science Hall

Format: 45-minute talk followed by approximately 15 minutes of Q&A

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Purpose and narrative

The talk asks how environmental processing changes the local gas cycle inside galaxies. It introduces MAUVE and selected results from the collaboration, then uses the resolved metallicity-SFR relation as the scientific centrepiece. A longer explanation of the gas-regulator framework connects that result to ongoing work on where star formation remains during ram-pressure stripping, internal metallicity structure, and molecular-gas tests.

The order follows the latest Yonsei discussion in [Seminar Package Advice](#). The accepted broad title is retained. The KIAS title and narrower emphasis are not used here. The timing is 45 minutes of speaking; questions are additional.

Central message: MAUVE connects environmental transformation to the local balance of gas supply, star formation and chemical enrichment. The observed metallicity-SFR coupling depends on local stellar surface density and metallicity diagnostic; the regulator offers a physical interpretation that can be tested with spatial structure and molecular gas.

Timing at a glance

Section	Slides	Duration	Clock	Role in the argument
Motivation: environmental evolution on resolved scales	1-3	4:00	00:00-04:00	Establish the physical question and the spatial scale
MAUVE and selected collaboration results	4-8	7:00	04:00-11:00	Show why a multiphase survey is needed
Paper I: metallicity and star formation	9-16	13:00	11:00-24:00	Present the evidence, spatial behaviour and diagnostic limits
The resolved gas-regulator interpretation	17-22	8:00	24:00-32:00	Explain supply, enrichment and finite response time
Ongoing work: where star formation remains	23-28	8:00	32:00-40:00	Distinguish SF occupancy, non-detection and detected non-SF emission
Spatial structure and direct molecular-gas tests	29-32	4:00	40:00-44:00	Define concrete next measurements
Conclusions and discussion	33	1:00	44:00-45:00	Reconnect environment, gas cycling and enrichment

The main presentation contains 33 slides. Six additional slides support questions and are outside the 45-minute budget. Slides are paced by the argument and by figure explanation, rather than by a fixed one-slide-per-minute rule.

1. Motivation: 00:00-04:00

Slide 1. Title and introduction - 0:30

Use the accepted title, presenter, affiliation and Yonsei date, with an existing MUSE galaxy image. Acknowledge Luca Cortese, Barbara Catinella, Luke Davies and the MAUVE collaboration verbally and in the notes. The opening is about the wider survey and physical problem, rather than only the paper title.

Slide 2. Environmental processing changes the local gas cycle - 1:30

Introduce gas removal, redistribution and the survival of inner gas. Frame the question as what happens to star formation and enrichment inside the remaining disc. Use an existing resolved galaxy image/map to make the spatial question concrete. Explain that fewer SF regions and weaker surviving SF regions are different observational outcomes.

Slide 3. Virgo at approximately 100 pc resolution - 2:00

Show the representative MUSE RGB fields and identify the diversity of discs, dust and structure. Explain spatial sampling versus effective resolution: 0.2 arcsec pixels sample the data, whereas seeing and the analysis binning set the approximately 100 pc measurement scale. The nine displayed fields are an illustration, not the survey sample size.

Transition: Resolving those structures is useful only if the gas phases and stellar populations can be connected.

2. MAUVE and selected collaboration results: 04:00-11:00

Slide 4. MAUVE connects gas and stars - 1:30

Expand the acronym: Multiphase Astrophysics to Unveil the Virgo Environment. Describe the 40-galaxy survey design, MUSE ionized gas and stellar spectroscopy, ALMA molecular gas, and complementary HST stellar/cluster information. Existing atomic-gas observations add wider environmental context. Distinguish programme design from the available subset used in an individual analysis. [S1]

Slide 5. A sample spanning environmental histories - 1:30

Explain pre-peak, close-to-peak and post-peak as an environmental ordering informed by gas morphology and dynamical context. This is a comparison of different galaxies, not repeated observations of the same galaxies through time. Keep the emphasis on the survival of cold gas in the inner disc, rather than promising a precise clock for every target. [S1]

Slide 6. NGC4383: an outflow with a cold-gas counterpart - 1:15

Use the existing MAUVE multiphase panels. Introduce the ionized outflow established by Watts et al. and the subsequent ALMA/MeerKAT study. Cortese et al. favour a fountain interpretation, with important uncertainty in separating cold outflowing gas from the extended warped disc. Do not equate an ionized outflow with permanent loss of the entire cold reservoir. [S2, S3]

Slide 7. NGC4064: feedback after stripping - 1:15

Use the existing Attwater et al. figure to illustrate an ionized bipolar outflow associated with centrally concentrated residual star formation in a stripped galaxy. Its role is to show how feedback and environmental processing can act together; the single case is not a universal evolutionary template. [S4]

Slide 8. Cluster discs show different ionization conditions - 1:30

Show the Brown et al. BPT comparison and explain the enhanced contribution of non-HII emission in the early MAUVE sample relative to PHANGS-MUSE field discs. The paper emphasizes the prominence of DIG as star formation declines, with an additional shock-like component. Elevated line ratios alone do not demonstrate widespread direct shock excitation by ram pressure. [S5]

Transition: These results make it essential to select the SF-dominated emission carefully before interpreting metallicity and SFR together.

3. Paper I: metallicity and star formation: 11:00-24:00

Slide 9. The local link between stars, star formation and metals - 1:00

Define stellar mass surface density, SFR surface density and gas-phase oxygen abundance. Briefly distinguish the resolved SF main sequence, the resolved mass-metallicity relation and the debated secondary SFR dependence. Use the EAS conceptual context figure, with its original attribution. [S6, S7]

Slide 10. Selecting star-forming emission - 1:30

Show the original BPT diagnostic figure and state the paper's HII selection: secure HII classification in both BPT diagrams, $EW(H\alpha) > 6$ Angstrom, and intrinsic Halpha dispersion < 45 km/s. SFR is derived from dust-corrected Halpha. The analysis uses 14 galaxies from internal release v2; ensemble relations exclude metallicity outlier NGC4383 and therefore contain 13 galaxies. Do not import the incorrect dispersion inequality printed in the old EAS slide. [S6]

Slide 11. The resolved mass-metallicity relation - 1:30

Show the original O3N2-M13 and O3N2-C20 relation panels. Explain the primary increase and high-density flattening before adding the SFR dependence. Identify stellar surface-density units explicitly. The two O3N2 recipes share an index and are not independent spectral diagnostics. [S6, Figure 2]

Slide 12. At fixed stellar density, the SFR dependence changes - 2:00

Use the original SFR-stratified relation and explain the low-density and high-density regimes separately. Higher local SFR is associated with lower O3N2-inferred abundance at low stellar surface density and with higher abundance at high surface density. Describe a conditional association, not an observed temporal response. [S6, Figure 3]

Slide 13. The sign changes near log stellar surface density 7.5-8.0 - 2:00

Show the Spearman-versus-density figure with its original galaxy colours and the black ensemble reference. Define the sign of the coefficient. State the approximate transition in $\log_{10}(\Sigma_{\text{star}} / [\text{solar masses kpc}^{-2}])$; it is not an exact universal threshold. This figure supplies the quantitative version of the preceding visual result. [S6, Figure 5]

Slide 14. Opposite local behaviours coexist in one disc - 1:30

Use the original NGC4192 offset and Moran-like map. Explain that correlated and anti-correlated spatial patterns can occur within one galaxy. The map is a local association diagnostic; neither its colour nor its absolute index is by itself a causal measurement or a calibrated significance map. [S6, Figure 4]

Slide 15. Metallicity calibration changes the inferred relation - 2:00

Compare the four diagnostics in an editable summary: O3N2-M13 and O3N2-C20 show the coherent inversion; D16 is mostly mildly positive across much of the density range; S-calibration PG16 is more heterogeneous and can show the opposite ordering. The primary rMZR is more stable than the secondary relation. Discuss ionization, N/O, DIG and shared line measurements as relevant systematics. Keep this caveat in the main talk. [S6, Section 5.1 and Figure 6]

Slide 16. What the observations establish - 1:30

Summarize the O3N2-dependent sign reversal and the coexistence of local behaviours. Separate the observed association from the question of its physical cause. Environment may modulate the gas cycle, but the Paper I result alone does not establish ram-pressure stripping as the cause of the inversion. [S6]

Transition: A useful model should explain how both signs can arise locally and identify the additional quantity we need to measure.

4. The resolved gas-regulator interpretation: 24:00-32:00

Slide 17. A local molecular reservoir - 1:00

Use the existing EAS reservoir illustration, clearly as a schematic. Define the local molecular gas reservoir, its gas supply, star formation and removal. Local supply means replenishment from the surrounding ISM, not automatically pristine accretion from outside the galaxy. The reservoir boundary matters. [S6, S7, S10]

Slide 18. Gas continuity - 1:30

Introduce the gas budget, with all rates per unit area:

$$\Sigma_{\text{SFR}} = \frac{\Sigma_g}{\tau_{\text{dep}}} ,$$

$$\frac{d\Sigma_g}{dt} = \Sigma_\phi - (1 - R)\Sigma_{\text{SFR}} - \Sigma_{\text{out}} .$$

Walk through supply, long-lived stellar locking and removal. State that instantaneous recycling and a single effective reservoir are model assumptions. Removal can include feedback and environmental effects, but the equation does not measure either rate from the MUSE maps. [S6, Section 4]

Slide 19. Metal enrichment competes with dilution - 1:30

Use the abundance equation:

$$\frac{dZ}{dt} = \frac{y(1 - R)\Sigma_{\text{SFR}} + (Z_\phi - Z)\Sigma_\phi}{\Sigma_g} .$$

Explain the positive enrichment term and the dilution term when supplied gas has lower abundance. Here Z is a linear abundance, not the plotted $12 + \log_{10}(\text{O}/\text{H})$. The explicit removal term cancels only under the assumption that removed gas has the reservoir abundance; removal still changes the reservoir, SFR and response time. [S6, Section 4]

Slide 20. The instantaneous target abundance - 1:30

Introduce the target and relaxation time:

$$Z^{\dagger} = Z_{\phi} + y(1 - R) \frac{\Sigma_{\text{SFR}}}{\Sigma_{\phi}},$$

$$\tau_{\phi} = \frac{\Sigma_g}{\Sigma_{\phi}}, \quad \dot{Z} = \frac{Z^{\dagger} - Z}{\tau_{\phi}}.$$

At fixed yield, recycling and supplied abundance, the first-order relation is

$$\delta Z^{\dagger} = (Z^{\dagger} - Z_{\phi})$$

$$\times (\delta \ln \Sigma_{\text{SFR}} - \delta \ln \Sigma_{\phi}).$$

Natural logarithms are used here. Supply rising proportionally more than SFR lowers the target; SFR rising proportionally more than supply raises it. This is a statement about the target under stated assumptions, not an unconditional theorem about the sign of an observed correlation. [S6]

Slide 21. The response has a finite timescale - 1:30

Explain that the observed abundance lags and smooths a changing target. A fixed-target response is used only as an analytic illustration. Correlations depend on how the drivers covary, their timescales and the phases sampled. A spatial map does not directly measure a time derivative. Do not identify the effective relaxation time uniquely with a measured turbulent-mixing time. [S6, Section 4.2]

Slide 22. Physical interpretation and a testable gap - 1:00

State the proposed interpretation: dilution/supply fluctuations can contribute to the low-density negative branch, whereas enrichment or depletion-time variations can contribute to the positive branch. MUSE does not independently measure the local supply rate. That gap motivates molecular-gas observations and simulation comparisons. [S6, S10]

5. Ongoing work: where star formation remains: 32:00-40:00

Slide 23. Where star formation remains along the RPS sequence - 1:00

Introduce the ongoing occupancy question. Present the analysis snapshot as 26 product-level systems after inclination and product-availability selection: 8 pre-peak, 5 close-to-peak and 13 post-peak.

NGC4567_8 is one combined product representing two physical components. These counts belong to this analysis, not the complete 40-target survey. [S8, S9]

Slide 24. SF, ND and NSF measure different outcomes - 1:30

Define SF as the selected HII/EW/dispersion population; ND as failure of the required Balmer detections; NSF as Balmer-detected emission that fails the SF selection. Define the denominator as the valid observed footprint. ND is sensitivity-limited, and NSF may mix DIG, shocks, old-star and nuclear excitation. The categories do not uniquely identify physical mechanisms. [S8]

Slide 25. SF occupancy is lower in the post-peak sample - 2:00

Use editable charts made directly from the existing equal-galaxy summary CSVs. Show the full valid footprint alongside the post-fit $S/N \geq 25$ subset. Full-footprint mean SF fractions are 0.388, 0.251 and 0.066 for pre-, close- and post-peak; corresponding ND fractions are 0.312, 0.509 and 0.727. Mark the result as preliminary and cross-sectional. Do not interpret the S/N cut as a neutral removal of bad data: it changes the denominator and preferentially excludes ND area. [S8]

Slide 26. Outer non-detections and central non-SF emission - 1:30

Show the existing equal-galaxy radial profiles, with cut and uncut curves explicitly identified. The descriptive picture combines a contracting SF footprint with a broader decline in occupancy; the destination differs with radius. Outer and low-density regions are increasingly ND, while detected NSF emission becomes prominent centrally. Avoid treating the stage average as the track of every galaxy. [S8, S9]

Slide 27. The maps show substantial galaxy-to-galaxy diversity - 1:30

Use selected existing spatial maps, including contrasting close-to-peak systems and a post-peak example. Keep SF/ND/NSF/excluded colours and labels unchanged. Connect this diversity to the small close-to-peak sample and the need for galaxy-level inference. Do not label an individual NSF feature as a confirmed shock or AGN using the category map alone. [S8, S9]

Slide 28. Occupancy and surviving-SF intensity are separate - 0:30

State the next measurement: compare the SFR of regions that remain SF at fixed stellar density and radius. The present fractions cannot tell whether those survivors are enhanced, normal or suppressed. [S8]

6. Spatial structure and molecular gas: 40:00-44:00**Slide 29. Spiral arms and HII-region structure - 1:00**

Introduce the planned work on small-scale metallicity fluctuations associated with spiral structure and the clumpiness of HII regions. Use an existing spiral image only to locate the question. Proposed measurements compare abundance residuals with arm/inter-arm position, HII-region structure and local gas; no arm-related metallicity detection is claimed here. [S0]

Slide 30. Spatial fluctuations as a physical test - 1:00

Explain a controlled comparison: remove broad stellar-density/radial trends, compare regions at matched physical resolution and selection, and examine coherence and offsets relative to the molecular and young-stellar structure. Simulations can test whether the inferred drivers reproduce the same observable relations after equivalent smoothing and selection. These are future tests. [S0, S6]

Slide 31. ALMA measures the reservoir and depletion time - 1:30

Define Sigma_H2 , $\text{tau_dep} = \text{Sigma_H2} / \text{Sigma_SFR}$ and molecular SFE. Explain the test of whether the positive branch associates with depletion-time changes or whether gas abundance and supply proxies are more informative. CO constrains molecular mass after conversion assumptions; it is not a direct map of the local replenishment rate. Match beams, apertures, sensitivity and tracer timescales. [S1, S10]

Slide 32. A connected research programme - 0:30

Connect the three ongoing directions: environmental SF occupancy, internal metallicity structure, and molecular-reservoir tests. Emission-line products and simulation comparisons support diagnostic robustness and physical interpretation; avoid making them a separate long project list. [S0]

7. Conclusions: 44:00-45:00

Slide 33. Three messages to take into the discussion - 1:00

1. MAUVE resolves how gas, star formation and enrichment coexist within environmentally processed Virgo galaxies.
2. The main O3N2-based result is a stellar-surface-density-dependent reversal of the metallicity-SFR association; its diagnostic dependence is part of the result.
3. Environmental occupancy, internal structure and molecular gas provide complementary tests of the local regulator interpretation.

Close with the question of which local conditions determine whether replenishment or enrichment dominates. Acknowledge the collaboration and invite discussion.

Backup slides and likely questions

Backup	Subject	Use during questions
B1 / 34	Survey and analysis sample bookkeeping	40 survey targets, 14 Paper I galaxies, 13 in the ensemble relation, 26 product systems in the ongoing occupancy snapshot
B2 / 35	Full metallicity-diagnostic comparison	Original four-diagnostic figure; calibration-dependent sign and scatter
B3 / 36	Metal continuity and the removal term	Derivation, well-mixed removal assumption and linear abundance convention
B4 / 37	Selection and uncertainty in the occupancy comparison	Cut versus uncut denominator; equal-galaxy weighting; whole-galaxy bootstrap values from the existing report
B5 / 38	What CO can and cannot measure	Molecular mass, depletion time, supply-rate degeneracy and conversion assumptions
B6 / 39	References and related MAUVE results	Paper links and source details for follow-up

Delivery and rehearsal guidance

The slides use existing scientific figures and images with source citations in speaker notes. New summary tables and charts remain editable. The notes contain slide time budgets, cumulative checkpoints, explanation cues, transitions and scientific limits.

Rehearse to the 11-, 24-, 32- and 40-minute checkpoints. If running late, shorten the two outflow examples by 30 seconds each, compress slide 14 by 30 seconds, and combine the two spatial-structure slides verbally. Preserve the diagnostic caveat, the occupancy definitions and the closing synthesis. This offers approximately two minutes of recovery without changing the scientific argument.

The stage results are an existing September analysis snapshot and have not been recomputed for this talk. The new work uses those stored figures and CSV values. Paper I and the survey results are sourced from their papers. The talk does not claim newly executed science pipelines, a measured local gas-supply-rate map, or new spiral-arm metallicity results.

Sources and figure provenance

All local paths below are relative to /Users/Igniz/Desktop/ICRAR/.

S0 - Talk brief: [Seminar Package Advice](#), especially the latest user revision of the Yonsei scope and title. The quoted invitation specifies the 45-minute talk and 15-minute discussion on 22 September.

S1 - Survey: Catinella et al. (2025), *Multiphase Astrophysics to Unveil the Virgo Environment (MAUVE)*, The Messenger 195, 15-18. [DOI](#). Local: EAS2026/pdfs/Catinella2025.pdf; survey material also in EAS2026/EAS2026 SS7 Rongjun Huang.pptx.

S2 - Ionized outflow: Watts et al. (2024), *MAUVE: a 6 kpc bipolar outflow launched from NGC4383, one of the most HI-rich galaxies in the Virgo cluster*. [Paper](#). Local: EAS2026/pdfs/Watts2024.pdf.

S3 - Cold outflow: Cortese et al. (2026), *MAUVE: Cold neutral gas in the outflow of NGC4383 and evidence for a fountain flow*. [Author version](#). Local: EAS2026/pdfs/Cortese2026.pdf and EAS2026/panels/Cortese2026.png.

S4 - Stripped-galaxy feedback: Attwater et al. (2026), *MAUVE-MUSE: A Star Formation-driven Outflow Caught in the Act of Quenching the Stripped Virgo Galaxy NGC4064*. [Author version](#). Local: EAS2026/pdfs/Attwater2026.pdf and EAS2026/panels/Attwater2026.png.

S5 - Ionization and kinematics: Brown et al. (2026), *MAUVE-MUSE: Ionization and Kinematic Signatures of Environmental Effects on Virgo Cluster Disks*. [Author version](#). Local: EAS2026/pdfs/Brown2026.pdf and EAS2026/panels/Brown2026.png.

S6 - Main result and model: Huang et al. (2026), *MAUVE-MUSE: When Metallicity Follows or Fights Star Formation - A Mass-Dependent Inversion in Virgo Galaxies*. [Author version](#). Local: EAS2026/pdfs/Huang2026.pdf; manuscript MAUVE/MAUVE Rongjun Zgas-Sigma_SFR paper/main.tex. Final figure crops come directly from the local paper PDF, retaining its figure versions and numbering. Older standalone figures in extended/Plot_Paper_1/ were inspected but are not substituted for the paper figures.

S7 - Existing talk material: EAS2026/EAS2026 SS7 Rongjun Huang.pptx and EAS2026/Group MEETING Rongjun Huang.pptx. Existing reference images and reservoir illustration are reused with their provenance. The paper's correct dispersion selection overrides the old EAS typo.

S8 - Occupancy analysis snapshot: further/20260905_stage_SF_fraction_analysis.md and further/20260905_stage_SF_fraction_analysis_assets/. Numeric charts use stage_global_equal_uncut.csv and stage_global_equal_snr25.csv; other evidence uses the existing radial-profile, individual-map and bootstrap products. No science pipeline rerun is implied.

S9 - Latest scientific framing: MAUVE/20260906 Where Star Formation Remains along Infall Stages.md, with its referenced existing figures. Where its shorthand interpretation is stronger than the diagnostic evidence, the talk uses the explicit qualifications from S8.

S10 - Reservoir definitions:

MAUVE/20260831_Molecular_Gas_Supply_Definitions_and_Observational_Inference.md. Used to distinguish molecular mass, replenishment, transport and chemical conversion rates.

Image assets: further/20260907_MAUVE_seminar_poster_images/, particularly the representative-field mosaic and galaxy RGBs, with README.md and provenance.json. These are MUSE-derived colour composites; approximately 100 pc refers to effective resolution, not pixel sampling. The mosaic is cropped only for slide framing while keeping the displayed galaxy structures and their labels intact.

The [MAUVE publications page](#) was checked on 8 September 2026 for publication metadata. Dates and counts in older local files are retained only when explicitly attached to their corresponding analysis snapshot.